

Faiyaz Rahman

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TECHNICAL SKILLS

Languages: TypeScript, Python, Java, Scala, Go, SQL, Terraform, Helm

AI/ML: AWS Bedrock, RAG, Vector DBs, fastembed, BM25, Prompt Engineering

Frameworks: Vue, React 18, Nest.js, Spring, Ktor, Flask, FastAPI

Platforms & Databases: Docker, Kubernetes, ArgoCD, Kafka, Strimzi, Karapace, PostgreSQL, MongoDB, Redis, Elasticsearch

Clouds: AWS (EC2, S3, Lambda, EKS, RDS), Alibaba Cloud

Architecture: Microservices, Event-Driven, CQRS, GitOps, CI/CD

EXPERIENCE

Full Stack Developer

Apr 2025 — present

Premialab

Full-time

- **Hybrid Search Screener:** Built a sub-300ms hybrid search screener over 100k+ instruments. Evaluated **Qdrant** prototype vs **pgvector/HNSW**, chose **Postgres** for operational simplicity. Fused **Cohere v4 + BM25 sparse** via **RRE**. **ag-Grid** frontend with virtualized rows, multi-tenant filter scopes, saved views. P95 under 500ms cold, under 300ms warm.
- **AI-Driven No-Code Reporting:** Architected a modular CRUD persistence engine using **Vue.js**, **FastAPI**, **NestJS**, and **AWS Bedrock (Claude 3.5 Sonnet)**. Implemented a **RAG pipeline** over financial reports to stream automated commentary alongside 20+ analytics components with exact PDF layout parity.
- **Factor Comparison & Peer Group Analytics:** Engineered peer group workflows and factor comparison engines using **Python (Pandas/NumPy/Jupyter)** and **PostgreSQL** to compute volume-adjusted betas, rolling regressor graphs, and KDE distributions.
- **Client-Side Report Automation:** Eliminated a flaky, legacy headless browser infrastructure by architecture-shifting to a client-side **jsPDF** implementation. Removed unpredictable server-side rendering failures while achieving an 80% reduction in report generation time and ensuring exact **WYSIWYG** parity.
- **Enterprise Whitelabel Enablement:** Enabled tier-one institutional rollouts for global clients including Societe Generale, Royal Bank of Canada, and Schroders by building tenant whitelabel configurations and automating team onboarding workflows.
- **Developer Experience (DX) Overhaul:** Accelerated local and CI workflow speeds by migrating legacy **ESLint** configs to Rust-based **Oxlint** and integrating **SonarQube** for automated code review.

Junior Full Stack Developer

Jul 2023 — Feb 2025

KM.ON Asia Limited

Full-time

- **Multi-Cloud Data Residency:** Spearheaded the end-to-end architecture, development, and deployment of a full-stack **React.js** and **Express.js** energy platform ([news](#)) across **AWS (EU)** and **Alibaba Cloud (China)**. Owned the full lifecycle of **Terraform** IaC, **ArgoCD**, and **Helm** charts while independently engineering custom data-routing layers within the application to enforce strict **GDPR** and **PIPL** compliance.
- **Terabyte-Scale Aggregation:** Engineered a two-tier fan-out **Python/Lambda/FCompute** data pipeline to ingest over 1.5 TB of raw IoT log histories from S3/OSS into active relational storage. Implemented memory-capped streaming mechanics monitored via **Grafana/CloudWatch** to eliminate OOM failures.
- **Time-Series Database Scaling:** Deployed and maintained an **AWS RDS Aurora PostgreSQL** cluster extended with **TimescaleDB** hypertables partitioned by telemetry timestamps. Handled a sustained write-throughput of 2,000+ operations per second while preventing index bloat.
- **Zero-Downtime Event Streaming:** Executed a live production infrastructure migration in **Java 17** from legacy messaging layers to **Strimzi Kafka** and **Karapace**. Authored custom **KafkaConnectors** processing 10M+ messages daily with inline sink verification and Dead Letter Queues (DLQ) to ensure 100% data parity.
- **High-Throughput IoT Network:** Developed a **Scala** microservice running over **MQTT** to stream bidirectional binary BLOB configurations between APIs and textile machinery. Built an edge command hub with custom exponential back-off loops to guarantee delivery of proprietary KMSF payloads over unstable networks.
- **Security Supply Chain & GitOps:** Integrated programmatic **SBOM** generation and **Trivy** vulnerability scans into the CI/CD pipeline for 15+ microservices to enforce strict EU/CN compliance. Configured automated semantic versioning workflows that parsed Git commit hooks to dynamically tag and build Docker artifacts.

IT Strategist

Feb 2021 — Aug 2021

BNP Paribas CIB

Contract

- **Latency-Aware Sync Engine:** Engineered a distributed, fan-out data replication engine in **Python** configured to dynamically route payloads based on live server latencies and node clustering. Bypassed GIL limitations via multiprocessing to synchronize terabyte-scale log data across 200+ nodes.
- **Real-Time Analytics Pipeline:** Accelerated data delivery by designing an ingestion pipeline using **Kafka** and **Elasticsearch**. Optimized cluster indexing strategies to achieve sub-second dashboard refreshes for high-cardinality trading logs.
- **Compliant Deployment Orchestration:** Developed automated shell scripts and platform workflows to enforce the mandatory **Four-Eyes Principle**. Configured multi-party approval validation for staging-to-production deployments and integrated programmatic **systemd** health checks.
- **High-Throughput Packet Replay Engine:** Optimized a multi-threaded **C++** simulation system replaying network packets from .pcap histories into trading systems. Resolved thread-synchronization bottlenecks by replacing blocking request-count evaluation loops with non-blocking condition variables.

PROJECTS

Homelab

2025 — present

- Engineered a **Golang** CLI toolchain with age-encrypted secrets, prioritized Docker Compose orchestration, and wildcard TLS management for 20+ services from a single service registry.
- Designed a zero-exposure network using **Tailscale** mesh VPN with MagicDNS, **Pi-hole** wildcard DNS, and **Caddy** reverse proxy. All services accessible without forwarded ports.
- Built a multi-layer observability stack: **VictoriaMetrics** metrics, **Loki/Promtail** log aggregation, **Grafana** dashboards, **Gatus** health checks, and **Scrutiny** disk monitoring.
- Built an agent harness with pre-commit hooks (**ruff**, **mypy**, **pytest**), SHA256-pinned images, append-only changelog auditing, and age-encrypted secrets management.

EDUCATION

The Chinese University of Hong Kong

Hong Kong

Bachelor of Science in Computer Science

- **Concentration:** Database and Information Systems
- **Honors:** Full Ride Admission Scholarship, The Sunny Award
- **Activities:** English Debating Team, Google Developers Student Club